

ENTHUSIAST, LEADER & ACHIEVER COURSE

PHASE : ALL ENTHUSIAST, MLA, B, K, M, P, Q, R, MAZA, B, C, D, E, F, G, H, L, M, N, O, P, Q, R, U, V,

MAAZ, W, X, Y, MAPA, PB, SPARK PRO, LAKSHYA

TARGET : PRE MEDICAL 2025

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TEST DATE : 11-04-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	3	3	2	2	2	1	1	1	1	3	4	2	1	3	3	2	1	4	2	4	4	1	3	3	3	1	2	2	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	2	2	2	4	1	3	2	2	2	3	3	4	3	3	3	3	4	1	2	4	3	1	1	3	1	3	2	4	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	2	2	3	2	2	4	2	1	3	1	2	1	1	4	2	2	1	4	3	2	2	3	4	3	2	1	2	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	1	4	1	4	4	2	4	3	3	3	3	2	1	1	3	3	4	4	3	3	4	2	2	1	3	1	3	3	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	3	3	1	1	2	2	4	3	4	1	3	1	3	4	4	2	2	1	3	4	2	1	3	4	2	1	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	1	2	2	3	4	2	2	3	3	3	1	4	3	4	3	3	4	2	1	4	1	3	4	3	2	4	2	1	2

HINT - SHEET

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|--|---|
| <ol style="list-style-type: none"> Ans (4)
NCERT XI Pg # 73 Ans (3)
NCERT-XI, Pg. # 76 Ans (3)
NCERT-XI, Pg. # 72 Ans (2)
NCERT-XI, Pg. # 72 Ans (2)
NCERT XI Pg. No. # 66 Ans (2)
NCERT Pg. # 63 Ans (1)
NCERT XI Page No. 58,60,62,67 Ans (1)
XI-NCERT Page No. # 65, 66 Ans (1)
NCERT XII Pg.# 8, 9 Ans (1)
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NCERT (XII) Pg # 4, 5 (E), 24 (H) Ans (4)
NCERT XII Pg. # 14 Ans (2)
NCERT XII Pg # 80 Ans (1)
NCERT-XII, Pg. # 86 Ans (3)
NCERT XII Pg. No. # 80 Ans (2)
NCERT XII Pg. No. # 62 Ans (1)
NCERT XII Pg. # 61 Ans (4)
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NCERT XII Pg. No :- 88 (E), 97 (H) Ans (4)
NCERT XII Pg. # 97 |
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HS-1/8

22. **Ans (4)**
NCERT XII Pg.# 122
23. **Ans (1)**
NCERT-XII, Pg.# 117
24. **Ans (3)**
NCERT XII Pg. # 93
25. **Ans (3)**
NCERT XII Pg. # 81
26. **Ans (3)**
NCERT XII Pg. # 103, Para # 6.3.3
27. **Ans (1)**
NCERT XII Page No. 91
28. **Ans (2)**
NCERT-XII, Pg # 188
29. **Ans (2)**
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30. **Ans (3)**
NCERT-XII, Pg.# 152, 153, 155
31. **Ans (4)**
NCERT-XII, Pg. # 248, Fig. 144(a)
32. **Ans (2)**
NCERT XII Page No. 208
33. **Ans (3)**
NCERT XII Pg. # 212
34. **Ans (2)**
NCERT XII Pg # 206-207
35. **Ans (4)**
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36. **Ans (1)**
NCERT XII, Pg. # 223
37. **Ans (3)**
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38. **Ans (2)**
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39. **Ans (2)**
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40. **Ans (2)**
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41. **Ans (3)**
NCERT XII Pg. # 198
42. **Ans (3)**
NCERT XII Pg.# 201-202
43. **Ans (4)**
NCERT-XII, Pg. No. # 230
44. **Ans (3)** Telegram: @NEETxNOAH
NCERT XII Pg.# 229(E) ; 251(H)
45. **Ans (3)**
NCERT XII Pg. # 191
46. **Ans (3)**
NCERT XII Pg. # 198, Fig. 11.3
47. **Ans (3)**
NCERT Pg. # 200
48. **Ans (4)**
NCERT XII, Page # 199
49. **Ans (1)**
NCERT-XII, Pg. # 199
50. **Ans (2)**
NCERT XII Pg # 169
51. **Ans (4)**
NCERT (XII) Pg. # 172
52. **Ans (3)**
Bioreactors are used in biotechnology to process large volumes of culture for the production of desired products such as proteins, enzymes, antibiotics, and other metabolites. They provide a controlled environment for the growth of microorganisms or cells, enabling efficient large-scale production.
53. **Ans (1)**
NCERT Pg # 183-184
54. **Ans (1)**
NCERT XIIth Page - 211
55. **Ans (3)**
NCERT Pg. # 181
56. **Ans (1)**
NCERT (XII) Pg # 213
57. **Ans (3)**
NCERT Pg. No.-180-Pest Resistant Plants.
59. **Ans (4)**
NCERT-XII, Page # 199, Fig. 11.4
60. **Ans (1)**
A selectable marker is a gene introduced into a cell or organism along with the gene of interest. It allows for the identification and selection of transformed cells that have successfully incorporated the foreign DNA. The marker helps in eliminating non-transformants by providing resistance to an antibiotic or another selective pressure, ensuring that only the cells containing the desired DNA survive.
61. **Ans (3)**
NCERT Page - 45
62. **Ans (2)**
NCERT Page - 46

63. **Ans (2)**
NCERT Page - 47
64. **Ans (3)**
NCERT Page - 47
65. **Ans (2)**
NCERT Page - 47
66. **Ans (2)**
NCERT Page - 48
67. **Ans (4)**
NCERT Page - 48
68. **Ans (2)**
NCERT Page - 49
69. **Ans (1)**
The **polar bodies** are small cells that are formed during the process of oogenesis (female gamete formation). They are produced as a result of the unequal division of the oocyte during meiosis.
 • In females, the oocyte is a **XX** chromosome pair, meaning both sex chromosomes are **X**.
 • During meiosis I, the oocyte divides to form a large secondary oocyte and a smaller first polar body, both of which are **XX**.
 • Meiosis II occurs only after fertilization. The secondary oocyte divides to form a **mature ovum** (also **XX**) and a **second polar body**.
70. **Ans (3)**
NCERT Pg. # 33, 34, 35
71. **Ans (1)**
NCERT Pg # 60, 61
72. **Ans (2)**
NCERT = 44
73. **Ans (1)**
NCERT Pg no. 48
74. **Ans (1)**
Module 8 Pg # 104
75. **Ans (4)**
NCERT-XII, Pg.# 160
76. **Ans (2)**
NCERT-XII, Pg.# 158-159
77. **Ans (2)**
NCERT-XII Pg. # 147
78. **Ans (1)**
NCERT-XII Pg. No. 170 (H)
79. **Ans (4)**
NCERT-XII, Pg. # 158
80. **Ans (3)**
NCERT-XII, Pg # 132
81. **Ans (2)**
NCERT Pg#136,135
82. **Ans (2)**
NCERT XII Page No. # 134(E)/147(H)
83. **Ans (3)**
NCERT Pg#143
84. **Ans (4)**
NCERT XII Pg. No. # 143
85. **Ans (3)**
NCERT Pg. 47
86. **Ans (2)**
NCERT(XII) Pg# 135 Para: 7.6
87. **Ans (1)**
NCERT (XII) Pg#137/148(H) para 7.7
88. **Ans (2)**
NCERT-XII Pg# 139
89. **Ans (2)**
Coelacanth or lobefins evolved into the first amphibians that lived on both land and water.
90. **Ans (3)**
NCERT (XII) Pg. # 137,138
91. **Ans (2)**

$$v_d = \frac{eE}{m} \tau = \frac{eV}{m\ell}$$

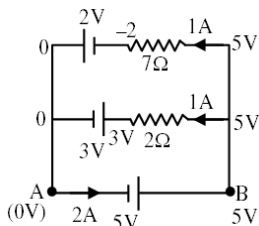
$$v_d \propto \frac{1}{\ell}$$

$$\frac{v_d}{v'_d} = \frac{2\ell}{\ell} = \frac{2}{1} \Rightarrow v'_d = \frac{v_d}{2}$$
92. **Ans (1)**

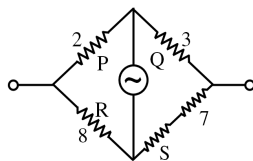
$$\Rightarrow \frac{R_v \times 400}{R_v + 400} = 300 \Rightarrow 4R_v = 3R_v + 1200$$

$$\Rightarrow R_v = 1200\Omega$$
93. **Ans (4)**

$$I = nqAV_d$$

$$J = \frac{I}{A} = nqV_d$$
 Direction of 'I' and 'J' are same.
94. **Ans (1)**


95. Ans (4)



$$8 \times 3 = 2(S + 7)$$

$$S + 7 = 12 \Rightarrow S = 5\Omega$$

96. Ans (4)

For maximum power across variable resistance, value of it must be equal to fix resistance of circuit.

97. Ans (2)

Induced charge on nearby surface of neutral conductor is of opposite nature so net force is attraction.

98. Ans (4)

$$\vec{a} = \frac{q\vec{E}}{m} = 24\hat{i} - 4.8\hat{j} \text{ m/s}^2$$

$$S_x = u_x + \frac{1}{2}a_x t^2 = \frac{1}{2} \times 24 \times (3)^2 = 108 \text{ m } (\because u_x = 0)$$

99. Ans (3)

$$r = \frac{kq}{V} = \frac{9 \times 10^9 \times 4 \times 10^{-9}}{30} \Rightarrow r = 1.2 \text{ m}$$

100. Ans (3)

$$\frac{1}{C_1} = \frac{1}{6} + \frac{1}{3} \Rightarrow C_1 = 2\mu\text{F}$$

$$C_{eq} = 2\mu\text{F} + 2\mu\text{F} = 4\mu\text{F}$$

$$U = \frac{1}{2}C_{eq}V^2 = \frac{1}{2}4(2)^2 = 8\mu\text{J}$$

101. Ans (3)

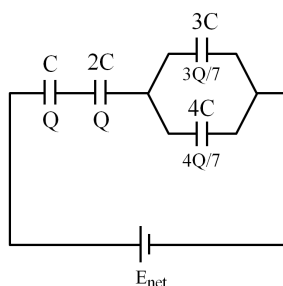
Insulator which can be polarised are used as dielectrics.

102. Ans (3)

Capacity does not depend on charge, but it depends on shape, size of capacitor and medium between plates.

103. Ans (2)

NCERT, Pg. # 72



$$V_1 = \frac{Q}{C}, V_2 = \frac{Q}{2C}$$

$$V_3 = V_4 = \frac{Q}{7C}$$

104. Ans (1)

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$$C = \frac{\epsilon_0 A}{d}$$

as d is increased, ' C ' decreases and so does charge q .

$$\therefore W_b = -\Delta qV$$

105. Ans (1)

Capacitance does not depend on charge on capacitor.

106. Ans (3)

$$(A) \vec{F} = q(\vec{V} \times \vec{B})$$

$$(B) F = qVB\sin\theta$$

since, $\sin\theta = 0$ here ($\theta = 0^\circ$)

So force = 0

107. Ans (3)

Lines of magnetic field around a current carrying wire are circular.

108. Ans (4)

If charges are of same nature then repulsion between them. If the move with same velocity then no magnetic force between them.

109. Ans (4)

$$M = NIA$$

Magnetic field.

110. Ans (3)

Dia-magnetic substances are repelled by external magnetic field.

111. Ans (3)

$$(a) \text{ Magnetic moment } M = I.A \Rightarrow \text{unit } A\cdot m^2$$

$$F = BI\ell$$

$$N = \left(\frac{Wb}{m^2} \right) A - m \text{ or } N = \left(\frac{Wb}{m^3} \right) A - m^2$$

$$\Rightarrow \frac{N - m^3}{Wb} = A - m^2$$

RHS is unit of magnetic moment so LHS also

$$(b) \text{ Magnetic induction } B = \frac{F}{I\ell} = \frac{N}{A - m}$$

$$(c) \phi = LI$$

$$B.A. = LI \Rightarrow (\mu_0 nI)A = LI$$

$$\text{Permeability } \mu_0 = \frac{L}{nA}$$

$$\text{unit of } \mu_0 = \frac{H}{(m^{-1})(m^2)} = H/m$$

(d) Intensity of magnetisation

$$H = nI \text{ or unit of } H = (m^{-1})A$$

unit of $H = A/m$

112. Ans (4)

Zero due to symmetry.

113. Ans (2)

$$X_C = \frac{1}{\omega C} = \frac{10^6}{10^2 \times 100} = 100 \Omega$$

$$Z = \sqrt{100^2 + 100^2} = 100\sqrt{2} \Omega$$

$$I_0 = \frac{V_0}{Z} = \frac{200\sqrt{2}}{100\sqrt{2}} = 2A$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = \sqrt{2}A$$

114. Ans (2)

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{10^2 + (2\pi \times 50 \times 20 \times 10^{-3})^2}$$

$$\sqrt{100 + 4\pi^2} = \sqrt{140}$$

$$Z = 11.8 \Omega$$

115. Ans (1)

I case :-

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + X_L^2}} \quad \text{where, } X_L = \omega L$$

In II case :-

$$L' = \frac{L}{2}$$

$$X_L' = \frac{\omega L}{2} = \frac{X_L}{2} \downarrow$$

So $Z' \downarrow$ & $I' \uparrow$

Glowness of bulb will increase

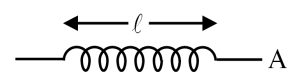
116. Ans (3)

$$q_{total} = -\frac{\Delta\phi_B}{R} = \frac{-\int_{t_1}^{t_2} e dt}{R} = \frac{-\int_0^2 (2t-3)dt}{2}$$

$$= \frac{-\left[2\frac{t^2}{2} - 3t\right]_0^2}{2} = \frac{-\left[\{(2)^2 - 3(2)\} - \{(0)^2 - 3(0)\}\right]}{2}$$

$$= \frac{-[-2] - \{0\}}{2} = \frac{+2}{2} = +1 C$$

117. Ans (1)



$$B = \mu_0 n I$$

$$\phi = \mu_0 n I N A = \mu_0 n^2 I \ell A$$

$$L = \mu_0 n^2 A \ell$$

$$U = \frac{1}{2} L I^2 = \frac{1}{2} \mu_0 n^2 A \ell I^2 = \frac{\mu_0^2 n^2 I^2}{2\mu_0} A \ell$$

$$U = \frac{B^2}{2\mu_0} A \ell$$

$$\text{Energy density} = \frac{B^2}{2\mu_0}$$

118. Ans (3)

$$L_{eq.} = L_1 + L_2 + 2M = 10 + 15 + 2 \times 2.5 = 30 H$$

119. Ans (3)

$$k = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{5000\pi} = 4 \times 10^{-4} m = 0.4 mm$$

120. Ans (3)

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 (I_c + I_d)$$

121. Ans (2)

$$\text{Expression } \bar{A}B + A\bar{B} + AB$$

$$= \bar{A}B + AB + \bar{A}B + AB \quad (\because A + A = A)$$

$$= B(A + \bar{A}) + A(B + \bar{B})$$

$$= B + A = A + B$$

122. Ans (3)

$$I_L = \frac{V_Z}{R_L} = \frac{10}{2 k\Omega} = 5 mA$$

$$I = \frac{V - V_Z}{R_S} = \frac{18 - 10}{250} = 32 mA$$

$$I_Z = I - I_L = 32 - 5 = 27 mA$$

123. Ans (3)

N-type semiconductor is electrically neutral.

124. Ans (3)

$$Y = (A + B) \cdot C$$

$$\text{for } Y = 1, A + B = 1 \text{ and } C = 1$$

$$\text{for } A + B = 1, A = 0, B = 0$$

125. Ans (1)

For convex mirror

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \quad \text{For } \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

Here f is positive for any real object u is negative

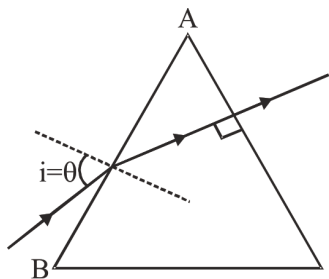
$$v = \frac{fu}{u-f} \quad \text{So v is positive}$$

$$m = -\frac{v}{u}$$

m is always positive image is always erect for real object.

126. Ans (1)

Given, $i = \theta$; $A = 5^\circ$; $\mu = 1.5$



For ray emerging normally from AC, $e = 0$

From relation, $A + \delta = i + e$, we have

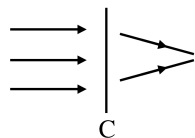
$$\Rightarrow 5^\circ + \delta = \theta + 0 \Rightarrow i = \theta = 5^\circ + \delta$$

Also, for small angles, $\delta = (\mu - 1)A$

$$\Rightarrow \delta = 2.5^\circ$$

$$\text{So, } i = \theta = 5^\circ + 2.5^\circ = 7.5^\circ$$

127. Ans (2)



So C is convex lens

128. Ans (2)

$$\sin \theta = \frac{\lambda}{a} \Rightarrow \sin \theta = \frac{7 \times 10^{-7}}{14 \times 10^{-7}} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

Angular width of central maxima $= 2\theta = 60^\circ$

129. Ans (4)

Say, intensity of unpolarised light is I_0

Then, according to question, intensity of final transmitted light.

$$I_{\text{out}} = \frac{I_0}{2} \cos^2 53^\circ$$

$$= \frac{I_0}{2} \times \left(\frac{3}{5}\right)^2 = \frac{9I_0}{50}$$

So, in terms of percentage of incident light

$$\% I_{\text{out}} = \frac{9}{50} \times 100 = 18\%$$

130. Ans (3)

Max. energy of emitted X-photon = Energy of bombarding electron

$\therefore$ Max. momentum of X-photon

$$= \frac{E}{c} = \frac{30 \times 10^3 \times 1.6 \times 10^{-19}}{3 \times 10^8} = 1.6 \times 10^{-23} \text{ N-s}$$

131. Ans (4)

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$$n = n_1 + n_2 = \frac{I_1 A \lambda_1}{hc} + \frac{I_2 A \lambda_2}{hc} = \frac{A}{hc} \left(\frac{I_1}{2} \lambda_1 + \frac{I_2}{2} \lambda_2 \right)$$

$$\approx 5 \times 10^{24} \text{ J}^{-1} \text{ m}^{-1} \times \frac{3.6 \times 10^{-3}}{2} (4000 + 5000) \times 10^{-10}$$

$$\frac{N}{t} = n = 5 \times 1.8 \times 9 \times 10^{24-3-7} = 81 \times 10^{14} \text{ s}^{-1}$$

$$N = 81 \times 10^{14} \times 60 (\text{min.})^{-1} = 4.86 \times 10^{17} (\text{min.})^{-1}$$

132. Ans (1)

Same ϕ , therefore from photo electric equation

$$eV_0 = (hc/\lambda) - \phi$$

from the graphs, it is clear that $(V_0)_1 > (V_0)_2$

$$\Rightarrow v_1 > v_2 \Rightarrow \lambda_1 < \lambda_2$$

$$\text{And } (i_s)_2 > (i_s)_1$$

133. Ans (3)

$$\lambda \propto \frac{1}{\sqrt{KE}}$$

$$KE_i = K$$

$$KE_f = (100\% - 19\%) KE_i$$

$$\frac{\lambda_f}{\lambda_i} = \sqrt{\frac{KE_i}{KE_f}} = \sqrt{\frac{K}{0.81K}} = \frac{1}{0.9} = \frac{10}{9}$$

$$\text{Now, } \frac{\Delta \lambda}{\lambda} \times 100\% = \left(\frac{\lambda_f}{\lambda_i} - 1 \right) \times 100\%$$

$$= \left(\frac{10}{9} - 1 \right) \times 100\% = 11.11\%$$

134. Ans (1)

$$K_{\text{max}} = h\nu - h\nu_0 = h\nu - \phi_0 \therefore \text{For emission}$$

$h\nu > \phi_0$ is essential

135. Ans (3)

$$\therefore K_{\text{max}} = h\nu - \phi_0 = h\nu - h\nu_0$$

$$\therefore K_{\text{max}} = h(\nu - \nu_0)$$

138. Ans (2)



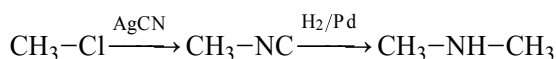
139. Ans (2)

Concept

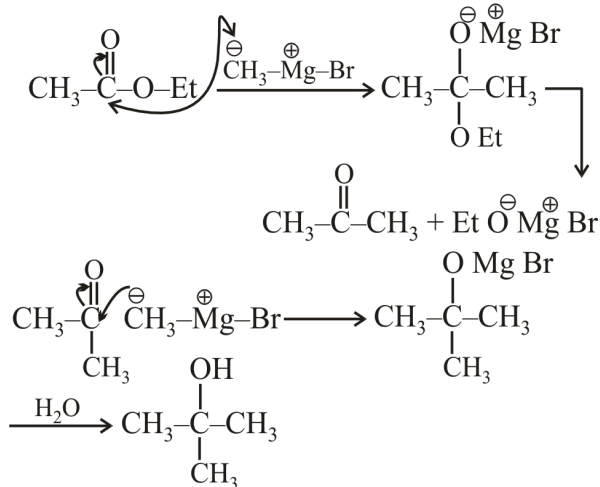
142. Ans (4)

NCERT, Pg. # 235-230

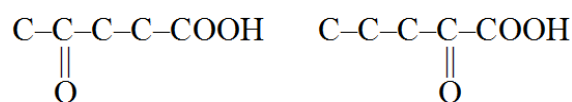
153. Ans (2)



155. Ans (3)



156. Ans (4)



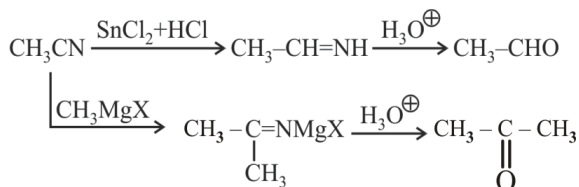
NaHCO₃ +Ve +Ve

DNP +Ve +Ve

AgNO₃/aq NH₃ -Ve -Ve

I₂/NaOH +Ve -Ve

158. Ans (2)



162. Ans (1)

Leaving group tendency $\propto$ Stability of leaving group

163. Ans (4)

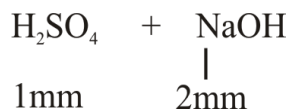
Propane - 1 - amine & Propane - 2 - amine, both are 1° Amine. Hence Cannot be distinguished by Hinsberg reagent.

164. Ans (3)

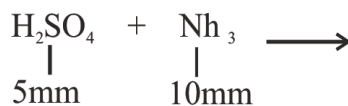
$$\begin{aligned}
 \% \text{ of chlorine} &= \frac{35.5 \times 0.287 \times 100}{143.5 \times 0.099} \\
 &= 71.72
 \end{aligned}$$

165. Ans (4)

$$\begin{aligned}
 \text{mole} &= \text{volume} \times \frac{\text{Normality}}{\text{v. f.}} \\
 \text{initial H}_2\text{SO}_4 &= 120 \times \frac{1}{10} \times \frac{1}{2} = 6\text{mm} \\
 \text{NaOH} &= \text{mole} = \text{volume} \times \frac{\text{Normality}}{\text{v. f.}} \\
 &= 20 \times \frac{1}{10} \times \frac{1}{10} = 2\text{mm}
 \end{aligned}$$



$$\text{H}_2\text{SO}_4 \text{ remaning} = 6\text{mm} - 1\text{mm} = 5\text{mm}$$



$$\begin{array}{c}
 \text{NH}_3 \Rightarrow \text{N}10 \rightarrow \text{mm} \\
 \downarrow \\
 10\text{mm} \\
 \text{mole}
 \end{array} \longrightarrow \text{mole N} = 10 \times 10^{-3}$$

$$\text{wt of N} = 10 \times 10^{-3} \times 14$$

$$\% \text{ of N} = \frac{10 \times 10^{-3} \times 14}{1.4} \times 100 = 10\%$$

166. Ans (3)

$$-\frac{d[\text{O}_2]}{dt} = \frac{9}{6} \frac{d[\text{CO}_2]}{dt}$$

167. Ans (3)

$$\begin{aligned}
 r_i &= k[\text{A}][\text{B}]^{1/3} \\
 r_f &= k \times \frac{[\text{A}]}{2} \times (8[\text{B}])^{1/3} = k[\text{A}][\text{B}]^{1/3} \\
 r_f &= r_i
 \end{aligned}$$

168. Ans (4)

$$\begin{aligned}
 t &= \frac{2.303}{k} \log \frac{A_0}{A_t} = \frac{2.303}{0.693/15} \log \frac{100}{10} \\
 &= 50 \text{ min}
 \end{aligned}$$

169. Ans (2)

$$\begin{aligned}
 \ln k &= \ln A - \frac{E_a}{RT} \\
 \text{Slope} &= \frac{-E_a}{R} = -2 \times 10^4 \\
 E_a &= (2 \times 10^4) (8.314 \times 10^{-3}) \\
 E_a &= 166 \text{ kJ/mol}
 \end{aligned}$$

170. Ans (1)

From slow step $r = k_3[\text{NOBr}_2][\text{NO}]$

from reversible step,

$$k_c = \frac{k_1}{k_2} = \frac{[\text{NOBr}_2]}{[\text{NO}][\text{Br}_2]}$$

$$[\text{NOBr}_2] = \frac{k_1}{k_2} \times [\text{NO}][\text{Br}_2]$$

$$r = k_3 \cdot \frac{k_1}{k_2} [\text{NO}]^2 [\text{Br}_2]$$

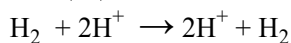
171. Ans (4)

$$\text{Number of equivalents of oxygen} = \frac{1.6}{8} = 0.2$$

$\therefore$ Number of equivalents of Ag deposited = 0.2

$$\text{Mass of Ag deposited} = 0.2 \times 108 = 21.6 \text{ g}$$

172. Ans (1)



$$P_1 \quad 1\text{M} \quad 1\text{M} \quad P_2$$

$$E_{\text{cell}} = 0 - \frac{0.059}{2} \log \frac{P_2}{P_1}$$

when $P_1 > P_2$ then $E_{\text{cell}} = +ve$

and $\Delta G = -ve$ (spontaneous)

173. Ans (3)

$$\begin{aligned} 2 \lambda_{\text{m}}^{\infty} \text{NH}_4\text{OH} &= 2 \lambda_{\text{m}}^{\infty} \text{NH}_4\text{Cl} + \lambda_{\text{m}}^{\infty} \text{Ba}(\text{OH})_2 - \lambda_{\text{m}}^{\infty} \text{BaCl}_2 \\ &= 2 \times 129.8 + 523.28 - 280 \\ &= 502.88 \end{aligned}$$

$$\lambda_{\text{m}}^{\infty} \text{NH}_4\text{OH} = 251.44 \text{ Scm}^2 \text{ mol}^{-1}$$

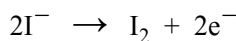
174. Ans (4)

NCERT-XII, Part-I, Pg # 89, 1st paragraph

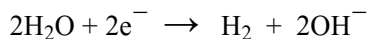
175. Ans (3)

aqueous KI solution

at anode



at cathode



$[\text{OH}^-] \uparrow$, $\text{pOH} \downarrow$

176. Ans (2)

$$m = \frac{M \times 1000}{1000d - MM_B}$$

177. Ans (4)

Telegram: @NEETxNOAH

$$\frac{P_B^0 - \frac{4}{5}P_B^0}{P_B^0} = \frac{n_A}{n_A + 10}$$

$$\frac{1}{5} = \frac{n_A}{n_A + 10}$$

$$n_A + 10 = 5n_A$$

$$4n_A = 10$$

$$\frac{w_A}{40} = 2.5$$

$$n_A = \frac{10}{4} = 2.5 \quad w_A = 40 \times 2.5 = 100 \text{ g.}$$

178. Ans (2)

$$P_P^0 = 70 \text{ torr}, X_P = 1 - X_Q = 0.8$$

$$P_{\text{total}} = P_P^0 \cdot X_P + P_Q^0 \cdot X_Q$$

$$84 = 70 \times 0.8 + P_Q^0 \times 0.2$$

$$\Rightarrow P_Q^0 = 140 \text{ torr}$$

179. Ans (1)

Henry's law - $P_{\text{gas}} = K_H \cdot X_{\text{gas}}$

$$2 = K_H \times 0.2 \quad \dots(i)$$

$$4 = K_H \times X_{\text{gas}} \quad \dots(ii)$$

$$\frac{2}{4} = \frac{0.02}{X_{\text{gas}}}$$

$$X_{\text{gas}} = 0.04$$

$$\therefore X_{\text{H}_2\text{O}} = 1 - 0.04 = 0.96$$

180. Ans (2)

For aqueous solution of glucose

$$\Delta T_f = k_f m$$

$$T_f^0 - T_f = k_f m$$

$$273.15 - 270 = k_f \times \frac{5/180}{95/1000} \quad \dots(1)$$

For aqueous solution of cane sugar

$$\Delta T_f = k_f m$$

$$273.15 - T_f = k_f \times \frac{5/342}{95/1000} \quad \dots(2)$$

(1) $\div$ (2) (Dividing)

We get, $T_f = 271.5 \text{ K}$ (F.P. of cane sugar solution)